Project proposal

Treatment Assistant and Field Logbook

The idea

On a small farm, applying plant protection products means doing calculations: the label dose comes either in g/HL or in g/ha, and the volume of spray mix per hectare depends on the nozzle and on how the sprayer is set. This project is an app to use next to the sprayer, which converts the label dose and tells how many tanks to prepare and how many grams to put in each. The farmer defines once their sprayers (tank capacity and a table of settings: nozzle, pressure and L/ha, as in the manufacturer's table), the products and the plots (for example, "Olival", an olive grove, with its area). Each calculation can be recorded as a treatment, building the field logbook of each plot. Calculated values must always be checked against the product label. I am interested in this project because I often help a small farm, where I have run into this problem: doing the calculations by hand every time is laborious, and an app that does them is much more practical.

Micro-narrative (PµFC, 34 words)

> *With the label in g/HL, Ana picks the product, the setting and "Olival", whose area is already defined. The app says how many tanks to prepare, the grams for each, and logs the treatment.*

Candidate entities and actions for classes and behaviors

Class and method names are kept in Portuguese, as in the code.

Classes

- **Produto** (product): the plant protection product, with the dose stated on the label.
- **Dose**: an abstraction with g/HL and g/ha variants. It represents only what the label says and receives the setting as a parameter when converting, without storing it, so the same product works with any sprayer or nozzle.
- **Pulverizador** (sprayer): tank capacity and list of settings.
- **Regulacao** (setting): nozzle, pressure and L/ha, composed inside the sprayer.
- **Parcela** (plot): name and area, with a stable identifier independent of the name; it is archived rather than deleted. "Olival" and "Vinha" (vineyard) are instances of Parcela, not classes.
- **PlanoDeTratamento** (treatment plan): the provisional process that happens before the treatment. It gathers the chosen product, sprayer, setting and plot, and owns the calculations.
- **Tratamento** (treatment): the record of a treatment already carried out, created from the plan when the farmer confirms. It is a snapshot of the moment and stores what was prepared (product, date, plot name and identifier, treated area, tank volumes, grams per tank and totals) and what justifies it (label dose with unit and the setting used: nozzle, pressure and L/ha). It does not store tank capacity (implicit in the tanks) nor equivalents in other units (these are viewed recalculated).

State

- **Stored:** plot name and area, sprayer capacity and settings, product dose and treatment history.
- **Derived** (always recalculated in the plan): total spray mix volume, number of tanks (including a last partial tank) and grams per tank.
- **History:** results are the truth of the record and inputs are context. Changing an area, a setting, a dose or a formula later does not rewrite history, and a past calculation error stays visible instead of being silently corrected.

Behaviors

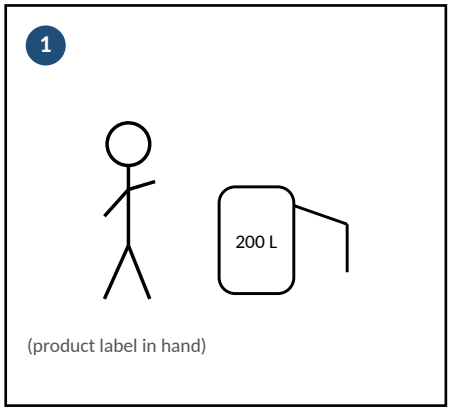
- Dose.converter_dose(regulacao)
- PlanoDeTratamento.calcular_volume_total, calcular_depositos, gramas_por_deposito and registar (creates the Tratamento)
- Parcela.historico (query of the treatments)

Worked example (to validate the model)

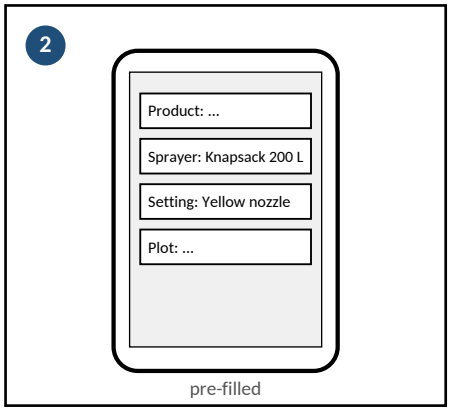
Olival of 2.5 ha, setting of 400 L/ha, 200 L tank, product at 150 g/HL: total volume = 2.5 × 400 = **1000 L**; tanks = 1000 / 200 = **5**; grams per tank = 150 g/HL × 2 HL = **300 g** (equivalent to 600 g/ha).

User stories as storyboards

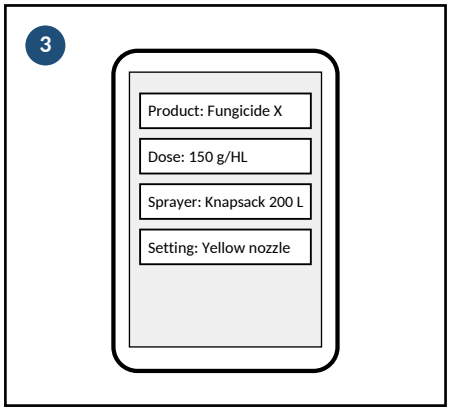
User story 1 *As a farmer, I want to choose the product, setting and plot, so I know how many tanks to prepare and how many grams to put in each.*



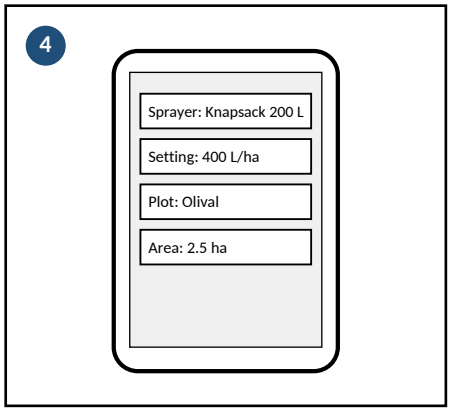
At the sprayer, the farmer has the phone and the product label.



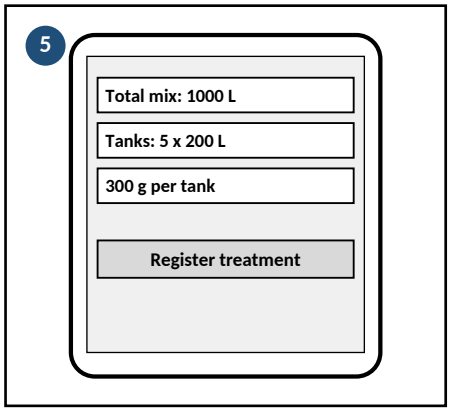
The app opens with the last sprayer and the last setting already chosen.



Picks the product; the app shows the label dose (150 g/HL).



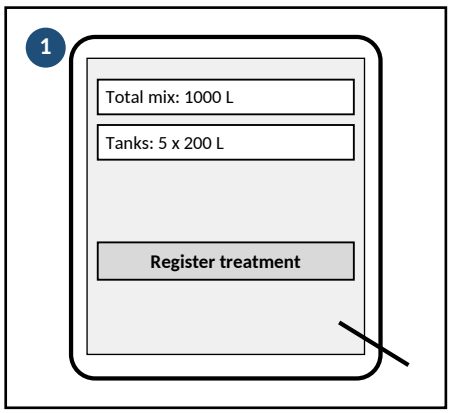
Picks the plot "Olival", whose area (2.5 ha) is already defined.



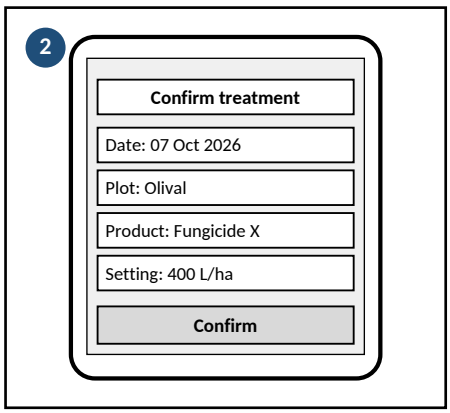
Result: 1000 L of spray mix, 5 tanks of 200 L and 300 g of product in each.

Example values; always check against the product label.

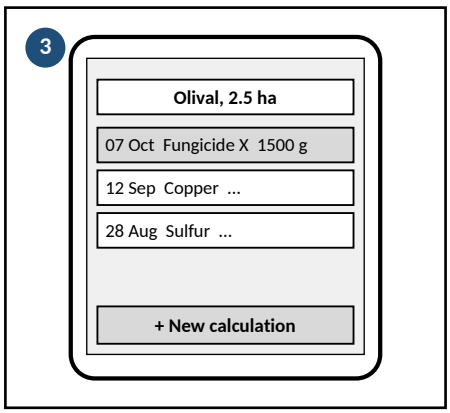
User story 2 *As a farmer, I want to record the treatment on the plot, so I have the history.*



On the calculated plan, taps "Register treatment" to turn it into a record.



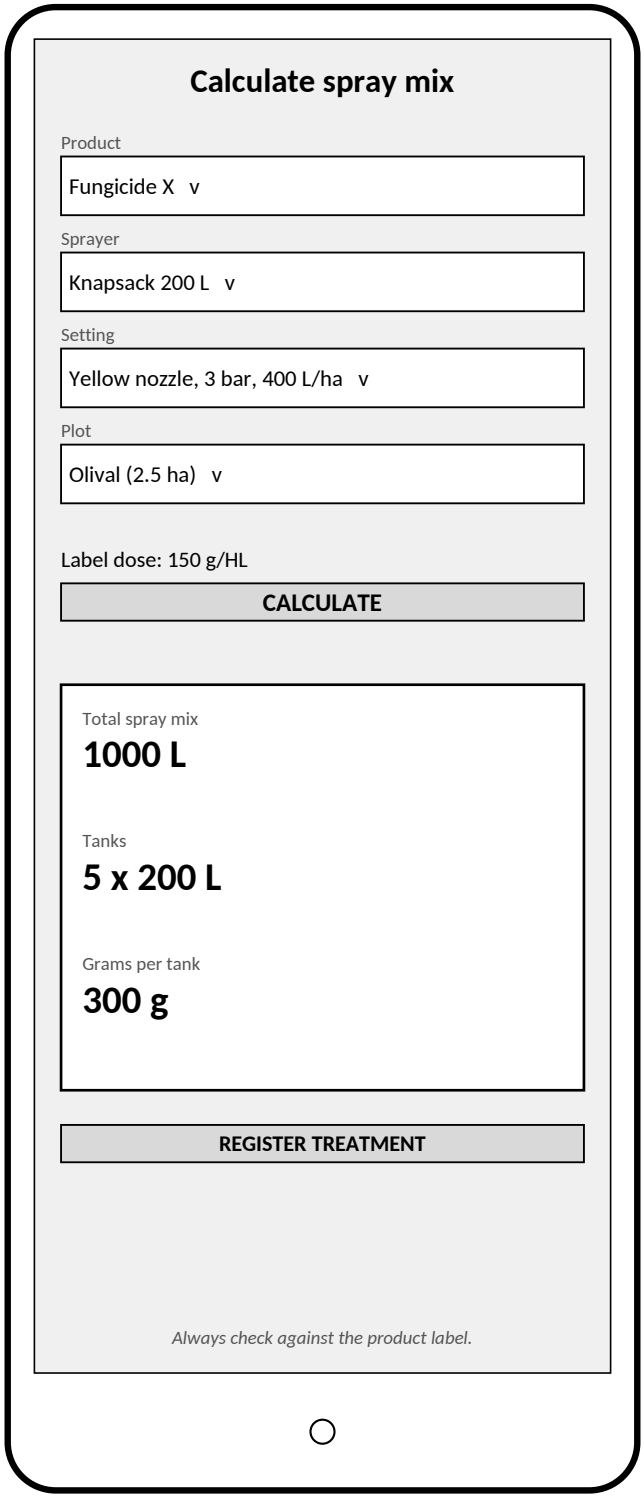
Confirms date, plot, product and setting used.



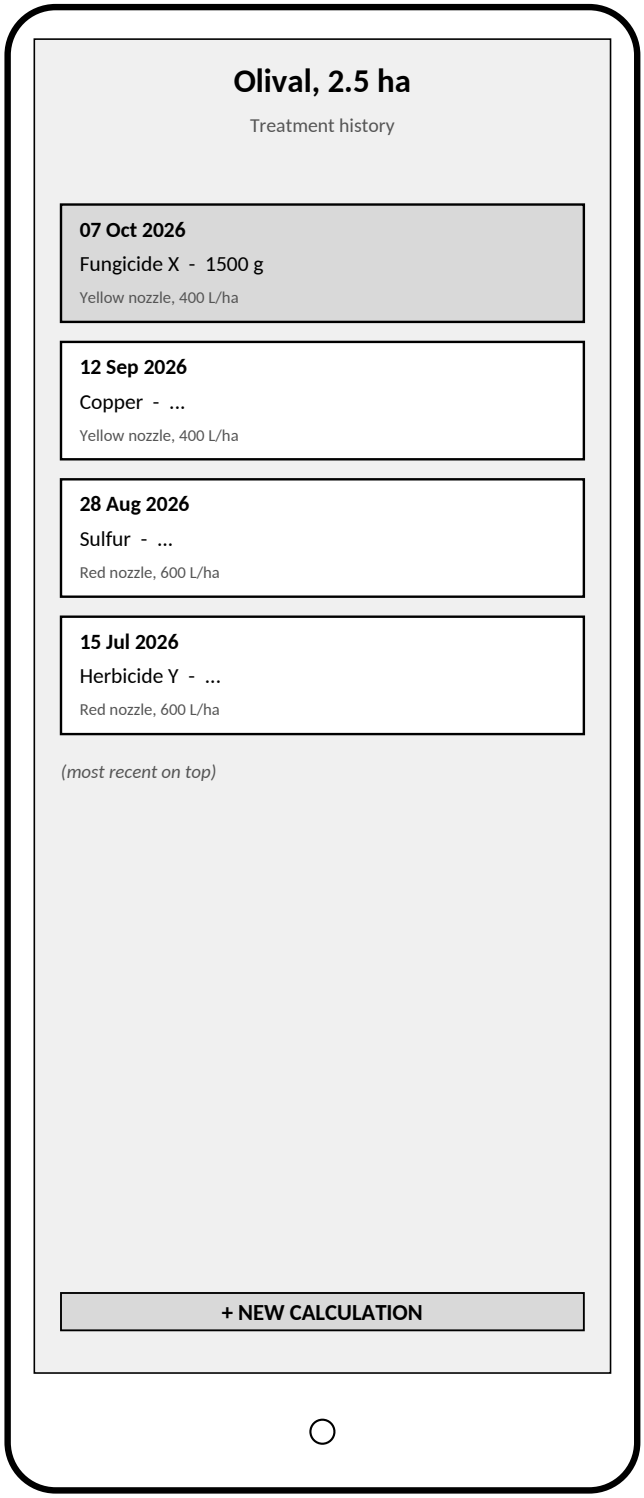
Sees Olival's history, with the new record at the top.

Low-fidelity mockups

Screen 1: Calculate spray mix



Screen 2: Plot history



"Register treatment" (screen 1) leads to screen 2; "+ New calculation" (screen 2) goes back to screen 1.